Tarea 9 (Ejercicio 3)

💚 Números complejos 💚

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Ejercicio 3. (100 puntos)

Si $z_1 = 1 - i$, $z_2 = -2 + 4i$, $z_3 = \sqrt{3} - 2i$, encuentra el valor de las siguientes expresiones:

i) $\left|\frac{z_1 + z_1 + i}{z_1 - z_2 - 1}\right|$

ii) $Im\left(\frac{z_1 z_2}{z_3}\right)$

iii) $Re\,(2z_1^2 + 3z_2^2 - z_3^2)$

Inciso i)

$$\left|\frac{\boldsymbol{z_1 + z_1 + i}}{\boldsymbol{z_1 - z_2 - 1}}\right| = \frac{\boldsymbol{\sqrt{493}}}{\boldsymbol{29}}$$

Resolución:

$$\left|\frac{z_1 + z_1 + i}{z_1 - z_2 - 1}\right| = \left|\frac{(1 - i\ ) + (-2 + 4i) + i}{(1 - i\ ) - (-2 + 4i) - 1}\right| = \left|\frac{1 - i\ - 2 + 4i + i}{1 - i + 2 - 4i - 1}\right| = \left|\frac{-1 + 4i}{2 - 5i}\right|$$

$$= \left|\left(\frac{-1 + 4i}{2 - 5i}\right)\left(\frac{2 + 5i}{2 + 5i}\right)\right| = \left|\frac{(-1 + 4i)(2 + 5i)}{(2 - 5i)(2 + 5i)}\right| = \left|\frac{-2 - 5i + 8i + 20i^2}{2^2 - (5i)^2}\right|$$

$$= \left|\frac{-2 + 3i + 20(-1)}{4 - (25)(-1)}\right| = \left|\frac{-2 + 3i - 20}{4 + 25}\right| = \left|\frac{-22 + 3i}{29}\right| = \left|-\frac{22}{29} + \frac{3i}{29}\right|$$

$$= \sqrt{\left(-\frac{22}{29}\right)^2 + \left(\frac{3}{29}\right)^2} = \sqrt{\frac{484}{841} + \frac{9}{841}} = \sqrt{\frac{493}{841}} = \frac{\sqrt{493}}{\sqrt{841}} = \frac{\sqrt{493}}{29}$$

Inciso ii)

$$\boldsymbol{Im\left(\frac{z_1 z_2}{z_3}\right)=\frac{4+6\sqrt{3}}{7}}$$

Resolución:

$$\frac{z_1 z_2}{z_3}=\frac{(1-i\,)(-2+4i)}{\left(\sqrt{3}-2i\right)}=\frac{-2+4i+2i-4i^2}{\sqrt{3}-2i}=\frac{-2+6i-4(-1)}{\sqrt{3}-2i}=\frac{-2+4+6i}{\sqrt{3}-2i}=\frac{2+6i}{\sqrt{3}-2i}$$

$$=\left(\frac{2+6i}{\sqrt{3}-2i}\right)\left(\frac{\sqrt{3}+2i}{\sqrt{3}+2i}\right)=\frac{(2+6i)\left(\sqrt{3}+2i\right)}{\left(\sqrt{3}-2i\right)\left(\sqrt{3}+2i\right)}=\frac{2\sqrt{3}+4i+6\sqrt{3}i+12i^2}{\sqrt{3}^2-(2i)^2}$$

$$=\frac{2\sqrt{3}+4i+6\sqrt{3}i+12(-1)}{3-4(-1)}=\frac{2\sqrt{3}+4i+6\sqrt{3}i-12}{3+4}=\frac{\left(2\sqrt{3}-12\right)+\left(4i+6\sqrt{3}i\right)}{7}$$

$$=\frac{2\sqrt{3}-12}{7}+\frac{\left(4+6\sqrt{3}\right)i}{7}$$

Ahora agarramos la parte imaginaria:

$$Im\left(\frac{z_1 z_2}{z_3}\right)=\frac{4+6\sqrt{3}}{7}$$

Inciso iii)

$$\boldsymbol{Re\left(2z_1^2+3z_2^2-z_3^2\right)=-35}$$

Resolución:

$$2z_1^2+3z_2^2-z_3^2=2(1-i\,)^2+3(-2+4i)^2-\left(\sqrt{3}-2i\right)^2$$

$$=2(1-2i+i^2)+3(4-16i+16i^2)-\left(3-4\sqrt{3}i+4i^2\right)$$

$$=2(1-2i+i^2)+3(4-16i+16i^2)-\left(3-4\sqrt{3}i+4i^2\right)$$

$$=2(1-2i-1)+3(4-16i-16)-\left(3-4\sqrt{3}i-4\right)$$

$$=2(-2i)+3(-12-16i)-\left(-1-4\sqrt{3}i\right)$$

$$=-4i-36-48i+1+4\sqrt{3}i=-35+\left(-52+4\sqrt{3}\right)i$$

Ahora agarramos la parte real:

$$Re\ (2z_1^2+3z_2^2-z_3^2)=-35$$